

Trevor Teolis

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Research Interests

Partial differential equations, kinetic theory, interacting particle systems, collective dynamics, sampling methods, operator learning, and mathematical foundations of machine learning.

Education

University of Illinois Chicago, Chicago, IL Ph.D., Mathematics, April 2025
Dissertation: *Well-posedness, long-time behavior, and multi-scale description of models of collective dynamics*. Advisor: Roman Shvydkoy.
University of Maryland, College Park, MD B.S., Mathematics and Computer Science, May 2019

Academic Positions

Rice University, Houston, TX Postdoctoral Researcher, Aug 2025–present
Advisor: Maarten V. de Hoop.
University of Illinois Chicago, Chicago, IL Doctoral Researcher, Aug 2020–May 2025
Advisor: Roman Shvydkoy.

Publications

1. R. Shvydkoy and T. Teolis. “Microscopic, mesoscopic, and macroscopic descriptions of the Euler Alignment System with adaptive communication strength.” *Discrete and Continuous Dynamical Systems*, 2025.
2. R. Shvydkoy and T. Teolis. “Well-posedness and long-time behavior of the Euler Alignment System with adaptive communication strength.” *Abel Symposium Proceedings*, 2024.
3. P. Nandori and T. Teolis. “Local Equilibrium of Particle Density in Planar Lorentz Processes.” *Nonlinearity*, 2021.

Preprints

1. T. Teolis, S. D. Mis, and M. V. de Hoop. “All Roads Lead to Markov Kernel Networks: A Representation Theorem for Discretization-Stable Learning.” Preprint, 2026.
2. T. Teolis and M. V. de Hoop. “In-context Maps as a Collision Operator: Stochastic Jump Transformer.” Preprint, 2026.
3. T. Teolis and M. V. de Hoop. “Target-adapted Green-Bessel SVGD: uniform-in-time propagation of chaos and last-iterate consistency.” arXiv:2609.08122, 2026.
4. S. Li, T. J. Maranzatto, J. Peszek, T. Teolis, S. Akkoc, K. Riedl, S. Ulukus, and N. García Trillos. “On the Diverse Dynamical Behaviors Arising in Deep Linear Transformers.” arXiv:2607.18584, 2026.
5. T. Teolis and M. V. de Hoop. “Riesz-Kernel Stein Variational Gradient Descent: Renormalized Entropy and Long-Time Particle Limits.” arXiv:2607.14527, 2026.
6. A. Balaji, T. Teolis, S. D. Mis, J. A. Lara Benitez, C. Wang, and M. V. de Hoop. “Hybrid operator learning of wave scattering maps in high-contrast media.” arXiv:2602.11197, 2026.
7. R. Shvydkoy and T. Teolis. “Long-time behavior of the Fokker–Planck–Navier–Stokes Alignment System.” arXiv:2508.07415, 2025.
8. A. Chertock, R. Shvydkoy, and T. Teolis. “Modulation of the monokinetic limit for models of collective dynamics.” arXiv:2508.05478, 2025.

Manuscripts in Preparation

1. T. Teolis, A. Siahkoohi, and M. V. de Hoop. “Finite-Mode Preconditioned SVGD with a Particle-Level Lyapunov Functional.”

Talks and Presentations

Invited Talks

- “Approximating Boltzmann solution maps by averaged measure-theoretic transformers,” minisymposium *Mathematical Foundations of Learning for Infinite-Dimensional Systems and PDEs*, SIAM Conference on Mathematics of Data Science, Salt Lake City, Utah, November 16–20, 2026. (upcoming)
- “Jump processes, stochastic in-context maps, mean-field limits, and the Boltzmann equation,” *Models of Emergence and Collective Dynamics*, 15th AIMS Conference, Athens, Greece, July 6–10, 2026.
- “Nanbu particles as stochastic in-context maps for Boltzmann dynamics,” *Measure Flows for Inverse Problems and Machine Learning*, SIAM Conference on Uncertainty Quantification, Minneapolis, Minnesota, March 22–25, 2026.

Contributed Talks

- Inverse Days 2026, Finnish Inverse Problems Society, University of Jyväskylä, Finland, December 14–17, 2026. (upcoming)
- “The s-model and the unconditional alignment of particles in a fluid,” Emergent Macroscopic Phenomena in Non-equilibrium Statistical Mechanics, GSSI, L’Aquila, Italy, May 2025.
- “A new model of collective dynamics with adaptive communication strength,” KinMAT Workshop, University of Warsaw, June 2024.
- “Proof techniques for alignment and flocking,” IDEAL Seminar (online), March 2024.

Research Presentations to Funders

- “FLOWERS: A warp drive for neural PDE solvers,” Oxy Annual Project Review Meeting, Houston, Texas, May 19, 2026.
- “Interacting particle systems in machine learning and inference,” TGS/GMIG Annual Project Review Workshop, Rice University, Houston, Texas, September 16, 2025.

Poster Sessions

- “The s-model and the unconditional alignment of particles in a fluid,” Workshop on Kinetic Theory and Fluids, UW–Madison, March 2025.
- “The s-model and the unconditional alignment of particles in a fluid,” Emergent Behavior in Complex Systems, IMSI, Chicago, March 2025.

Departmental Seminars

- “Long-time behavior and multi-scale description of models of collective dynamics,” Ph.D. Defense, UIC, April 2025.
- “A new model of collective dynamics with adaptive strength,” Analysis and Applied Math Seminar, UIC, February 2024.

Awards

- Victor Twersky Memorial Award, University of Illinois Chicago, 2024.
- Award for Graduate Research, University of Illinois Chicago, 2024.
- Travel Award, University of Illinois Chicago, 2024.

Workshops and Summer Schools

- Emergent Macroscopic Phenomena, GSSI, Italy, May 2025.
- Workshop on Kinetic Theory and Fluids, UW–Madison, March 2025.
- Emergent Behavior in Complex Systems, IMSI, Chicago, March 2025.
- Winter School: Boundary and Singularity in Fluid Mechanics, Stony Brook, January 2025.
- KinMAT Workshop, University of Warsaw, June 2024.
- Summer School: Fluid Dynamics, UMD, July 2023; Summer School: PDEs, Duke, August 2022.

Teaching

Rice University

- **CMOR 422/522: Numerical Analysis**, Co-Instructor, Fall 2026.
- **CMOR 677: Mathematical Theory of Deep Learning**, Co-Instructor, Spring 2026.

University of Illinois Chicago. Teaching Assistant for Calculus I–III, Differential Equations, and Linear Algebra, 2020–2025. Led discussion sections, mini-lectures, and office hours.

Professional Experience

Rampart Communications, Linthicum, MD

Developed and implemented polar-coded modulation schemes.

Summer Doctoral Intern, Jun–Aug 2024

Argonne National Laboratory, Lemont, IL

Developed algorithms for mixed-integer PDE-constrained optimization.

NSF Summer Intern, Jun–Aug 2021

National Institutes of Health, Bethesda, MD

Undergraduate Researcher and Software Engineer, Jan 2018–Jul 2020.

Built finite-element models of brain tumor growth using MRI and DTI data.